



Model Testing

Here is a guide on how to conduct model testing for a drone's model and algorithm effectively:

- 1. Define Scenarios:** Identify test scenarios reflecting different environmental and operational conditions (e.g., weather, terrain, altitude, wind speed).
- 2. Select Environments:** Choose test settings, including indoor, outdoor, urban, rural, and simulated environments.
- 3. Collect Data:** Gather relevant data during tests using sensors like GPS, LiDAR, cameras, and environmental sensors.
- 4. Design Test Cases:** Create test cases covering typical conditions, edge cases, and failure modes, specifying inputs, outputs, and success criteria.
- 5. Execute Flights:** Conduct test flights systematically, ensuring consistency and repeatability.



Alt text: A drone flying over a rural landscape



Alt text: A man inspecting a farm with a drone



- 6. Monitor Performance:** Track the drone's performance, focusing on perception, decision-making, obstacle navigation, and mission success.



Alt text: A person flying a drone

- 7. Evaluate Results:** Compare test outcomes with expected results, identifying strengths and weaknesses.
- 8. Identify Improvements:** Highlight limitations and propose enhancements to improve algorithm effectiveness.
- 9. Iterate and Refine:** Continuously refine models and algorithms based on test feedback and empirical data.
- 10. Document Findings:** Summarise the test process, results, and recommendations in detailed reports.